Task 3: Python for AI

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Topic: Python Programming Concepts with AI-Related Examples

1. INTRODUCTION

Python is a popular programming language used in Artificial Intelligence (AI), Machine Learning (ML), data science, and automation. Python helps developers create intelligent applications that can analyze data, make predictions, and solve real-world problems.

In this task, we will learn four basic Python concepts using AI-related examples:

- Variables
- Data Types
- User Input
- Operators

2. PYTHON PROGRAMS

Program 1: Variables – Student Pass/Fail Prediction

Aim: To use variables to store student marks and predict whether the student has passed or failed.

PYTHON CODE:

```
# Student Pass/Fail Prediction

student_name = "Yasasvi"
marks = 75
passing_marks = 35

print("Student Name:", student_name)
print("Student Marks:", marks)

if marks >= passing_marks:
    print("Prediction: PASS")
else:
    print("Prediction: FAIL")
```

OUTPUT:

```
Student Name: Yasasvi
Student Marks: 75
Prediction: PASS
```

Explanation:

- `student_name` stores the student's name.
- `marks` stores the student's marks.
- `passing_marks` stores the minimum passing marks.
- The program predicts PASS or FAIL using an if-else condition.

Program 2: Data Types – AI Student Information

Aim: To understand different Python data types using student information.

PYTHON CODE:

```
# Different Data Types

name = "Yasasvi"
age = 20
marks = 85.5
is_passed = True

print("Name:", name)
print("Age:", age)
print("Marks:", marks)
print("Passed:", is_passed)

print(type(name))
print(type(age))
print(type(marks))
print(type(is_passed))
```

OUTPUT:

```
Name: Yasasvi
Age: 20
Marks: 85.5
Passed: True

<class 'str'>
<class 'int'>
<class 'float'>
<class 'bool'>
```

Explanation:

- `str` – Stores text.
- `int` – Stores whole numbers.
- `float` – Stores decimal numbers.
- `bool` – Stores True or False values.
- These data types help AI systems process different kinds of information.

Program 3: User Input – AI Pass/Fail Prediction

Aim: To accept marks from the user and predict the student's result.

PYTHON CODE:

```python
# AI-Based Student Result Prediction

name = input("Enter student name: ")
marks = int(input("Enter student marks: "))

if marks >= 35:
    print(name, "is predicted to PASS")
else:
    print(name, "is predicted to FAIL")
```

OUTPUT:

```
Enter student name: Yasasvi
Enter student marks: 80

Yasasvi is predicted to PASS
```

Explanation:

- `input()` accepts information from the user.
- `int()` converts the entered marks into an integer.
- The program checks the marks and displays the predicted result.

Program 4: Operators – Weather Temperature Conversion

Aim: To use arithmetic operators to convert temperature from Celsius to Fahrenheit.

PYTHON CODE:

```python
# AI Weather Temperature Conversion

celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9/5) + 32

print("Temperature in Celsius:", celsius)
print("Temperature in Fahrenheit:", fahrenheit)
```

OUTPUT:

```
Enter temperature in Celsius: 30

Temperature in Celsius: 30.0
Temperature in Fahrenheit: 86.0
```

Explanation:

- `*` – Multiplication operator.
- `/` – Division operator.
- `+` – Addition operator.
- `=` – Assignment operator.

This program demonstrates temperature conversion, which can be used as a basic component in weather-related AI applications.

Program 5: Age Eligibility Checker

Aim: To check whether a person is eligible based on their age.

PYTHON CODE:

```
# AI-Based Age Eligibility Checker

name = input("Enter your name: ")
age = int(input("Enter your age: "))

if age >= 18:
    print(name, "is eligible.")
else:
    print(name, "is not eligible.")
```

OUTPUT:

```
Output 1:
Enter your name: Yasasvi
Enter your age: 20

Yasasvi is eligible.

Output 2:
Enter your name: Ravi
Enter your age: 16

Ravi is not eligible.
```

Explanation:

- `input()` accepts the person's name and age.
- `int()` converts the age into an integer.
- `if-else` checks the eligibility condition.
- If age is 18 or above, the person is eligible; otherwise, they are not eligible.

> **AI Application:** This is a basic example of rule-based decision-making that can be used in automated eligibility checking systems.

3. TYPES OF PYTHON OPERATORS

Operator	Example	Purpose
Arithmetic	a + b	Mathematical operations
Comparison	a > b	Compares values
Logical	a and b	Combines conditions
Assignment	a = 10	Assigns values

4. REAL-LIFE APPLICATIONS

1. **Student Performance Prediction**
   AI applications can analyze student marks and estimate academic performance.
2. **Weather Forecasting**
   AI systems analyze temperature and weather data to help predict weather conditions.
3. **Data Analysis**
   Python data types and operators help process information used in AI applications.

5. CONCLUSION

Through this task, I learned the basic concepts of Python programming, including variables, data types, user input, and operators.

I also developed simple AI-related programs for student pass/fail prediction, weather temperature conversion, and age eligibility checking.

These programs helped me understand how Python can be used to solve real-world problems and build a foundation for learning Artificial Intelligence and Machine Learning.